DTPA: Dynamic Token-level Prefix Augmentation for Controllable Text Generation

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Abstract

Controllable Text Generation (CTG) is a vital subfield in Natural Language Processing (NLP), aiming to generate text that aligns with desired attributes. However, previous studies commonly focus on the quality of controllable text generation for short sequences, while the generation of long-form text remains largely underexplored. In this paper, we observe that the controllability of texts generated by the powerful prefix-based method Air-Decoding tends to decline with increasing sequence length, which we hypothesize primarily arises from the observed decay in attention to the prefixes. Meanwhile, different types of prefixes including soft and hard prefixes are also key factors influencing performance. Building on these insights, we propose a lightweight and effective framework called **D**ynamic **T**oken-level **P**refix **A**ugmentation (**DTPA**) based on Air-Decoding for controllable text generation. Specifically, it first selects the optimal prefix type for a given task. Then we dynamically amplify the attention to the prefix for the attribute distribution to enhance controllability, with a scaling factor growing exponentially as the sequence length increases. Moreover, based on the task, we optionally apply a similar augmentation to the original prompt for the raw distribution to balance text quality. After attribute distribution reconstruction, the generated text satisfies the attribute constraints well. Experiments on multiple CTG tasks demonstrate that DTPA generally outperforms other methods in attribute control while maintaining competitive fluency, diversity, and topic relevance. Further analysis highlights DTPA's superior effectiveness in long text generation.

1 Introduction

The goal of Controllable Text Generation (CTG) is to generate text with desired attributes (e.g., sentiment, topic, non-toxic) (Prabhumoye, Black, and Salakhutdinov 2020; Liang et al. 2024). With the advent of large Transformer-based Casual Language Models (CLMs) such as GPT-2 (Radford et al. 2019) and LLaMA (Touvron et al. 2023), many methods based on them have achieved promising attribute control performance in this field. Despite this, it remains a persistent challenge to strike a good balance between attribute controllability and text quality.

In the early stages, CTG methods primarily focused on retraining all parameters of language models (Keskar et al. 2019; Chan et al. 2021; He 2021; Arora et al. 2022), which, despite being effective, incurred substantial computational costs. To improve efficiency, fine-tuning approaches (Zhang and Song 2022; Qian et al. 2022; Pei, Yang, and Klein 2023; Zhou et al. 2023) were proposed, leveraging additional fine-tuned adapters, prefixes, or prompts. While reducing trainable parameters, these methods remain resource-intensive and prone to overfitting on specific attribute datasets, limiting their generalizability. Another line of work (Liu et al. 2021; Yang and Klein 2021; Krause et al. 2021; Zhong et al. 2023; Wang and Demberg 2024) controls generation by intervening in the decoding stage, often adjusting output probabilities via attribute classifiers. These methods offer strong control effectiveness but are susceptible to Attribute Collapse (Zhong et al. 2023), where increasing the attribute control strength beyond a certain threshold leads to a decline in text quality.

In this work, we propose **D**ynamic **T**oken-level **P**refix **A**ugmentation (**DTPA**) for controllable text generation, a novel and effective framework built on the powerful prefix-based method Air-Decoding (Zhong et al. 2023). Our approach is motivated by two key observations.

First, we note that the prefixes used in Air-Decoding are learnable embeddings trained from attribute corpus, which are commonly referred to as soft prefixes. This raises the natural question: ***can natural language prompts (which we call hard prefixes), such as "A positive text:", achieve comparable or even better control?*** We design a set of hard prefixes and compare them with soft prefixes within the Air-Decoding framework. Results show that hard prefixes can outperform soft ones in tasks like sentiment control, motivating prefix-type selection tailored to specific attributes.

Second, while most CTG methods perform well on short text (e.g., max length 20 in PPLM (Dathathri et al. 2019) and DExperts (Liu et al. 2021), 50 in Air-Decoding (Zhong et al. 2023) and FreeCtrl (Feng et al. 2024)), ***their effectiveness in long text generation remains largely underexplored***, which is important for real-world applications such as story generation, dialogue systems, document-level summarization, and so on. Our preliminary study on Air-Decoding reveals that its attribute controllability degrades with increasing sequence length, likely due to prefix attention decay—a phenomenon consistently observed in our experiments. This insight motivates our augmentation of prefix attention to en-

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hance long-form controllability.

Building on the former two insights, we meticulously design our method DTPA. First, we select the optimal prefix type for the given CTG task. Then, we dynamically amplify the attention to the prefix for the attribute distribution using a length-dependent exponential scaling factor. Optionally, we apply a similar augmentation to the original prompt for the raw distribution to balance text quality. Finally, we reconstruct the attribute distributions to obtain a more attribute-aligned and fluent output. Extensive experiments on multiple CTG tasks demonstrate the effectiveness of DTPA.

To conclude, our main contributions are as follows:

1. Within Air-Decoding, we analyze the impact of different prefix types and observe that attribute controllability degrades with sequence length, along with prefix attention decay, which we hypothesize to be the main contributor.
2. Inspired by the insights above, we design DTPA based on Air-Decoding. It significantly alleviates the attention decay to the prefix, resulting in an output distribution that is more aligned with desired attributes.
3. Comprehensive experiments demonstrate the efficacy of DTPA in generating controllable text while maintaining competitive text quality, especially in long text generation, without requiring extensive hyperparameter tuning.

2 Related Work

Retraining. Retraining-based CTG methods modify the model architecture or parameters using large-scale attribute-specific data. Keskar et al. (2019) train a 1.63B-parameter conditional Transformer with 55 control codes. CoCon (Chan et al. 2021) injects control condition embeddings into hidden states via CoCon blocks with a self-supervised objective. POINTER (Zhang et al. 2020) is pre-trained with a progressive insertion-based objective on 12GB of Wikipedia and fine-tuned for hard-constrained generation tasks.

Prefix/Prompt-based. As CLMs scale up, retraining becomes computationally expensive. Prefix/prompt-based methods offer a lightweight alternative with minimal or no parameter updates. Prefix-Tuning (Li and Liang 2021) prepends trainable embeddings to each Transformer layer, while Prompt-Tuning (Lester, Al-Rfou, and Constant 2021) inserts them only at the input layer. Qian et al. (2022) train contrastive prefixes jointly, and Tailor (Yang et al. 2023) concatenates multiple soft prompts for multi-attribute control. PREADD (Pei, Yang, and Klein 2023) compares output logits obtained with and without attribute prompts.

Decoding-time Intervention. These methods control attributes by modifying the output distribution during inference, requiring minimal or no changes to model parameters. PPLM uses gradients from an attribute classifier to steer hidden states. FUDGE (Yang and Klein 2021) applies a Bayesian factorization to adjust token probabilities dynamically. DExperts combines outputs from expert and anti-expert models via contrastive decoding to guide generation. Air-Decoding constructs attribute distributions using prepended learned prefixes and reconstructs them to balance the weights of attribute-related and unrelated words, enhancing controllability while maintaining fluency.

Task	Attribute	Hard Prefix
Sentiment Control	Positive	`"Very positive:"`
	Negative	`"Very negative:"`
Topic Control	World	`"World-related:"`
	Sports	`"Sports-related:"`
	Business	`"Business-related:"`
	Science	`"Science-related:"`
Detoxification	Nontoxic	`"Very nontoxic:"`
	Toxic	`"Very toxic:"`

Table 1: The hard prefixes designed for each CTG task.

Task	Prefix	Acc / Tox.	PPL↓	Dist-1	Dist-2	Dist-3
Sent. Control	**Hard**	**97.40**	29.69	**0.14**	**0.57**	**0.79**
	Soft	96.82	**26.66**	0.13	0.55	0.78
Topic Control	Hard	80.30	31.58	0.07	0.46	0.73
	Soft	**97.21**	**31.18**	**0.08**	**0.47**	**0.74**
Detox.	Hard	26.3	50.45	0.07	**0.46**	**0.76**
	Soft	**18.5**	**49.29**	**0.07**	0.44	0.74

Table 2: Comparison between Air-Decoding using soft and hard prefixes. ↓ indicates lower is better. Sent. and Detox. stand for Sentiment and Detoxification, respectively.

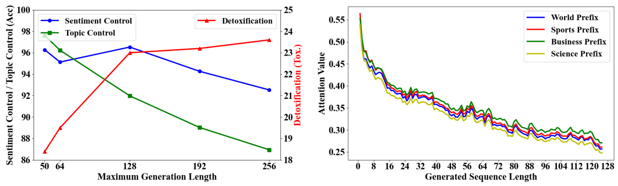
3 Insights

3.1 The Impact of Different Prefix Types

In Air-Decoding, the prefixes are learnable embeddings trained from attribute-specific datasets, which we call **Soft prefixes**. While soft prefixes generally offer precise guidance (Li and Liang 2021; Qian et al. 2022), their training overhead limits flexibility. Alternatively, natural language prompts—referred to as **hard prefixes** (e.g., "*A positive text:*")—can provide a lightweight option.

To assess their effectiveness, we replace the soft prefixes in Air-Decoding with manually designed hard prefixes on the three CTG tasks: Sentiment Control, Topic Control, and Detoxification. As shown in Table 1, the selected hard prefixes are short, semantically aligned with target attributes, and contain key attribute-related words, which can help the model focus on desired attributes with minimal distraction. For sentiment and detoxification, we include the degree adverb "*Very*", which, as shown in Appendix B, significantly improves attribute control, demonstrating the model's instruction-following ability. While for topic control, "*Very*" leads to a performance drop, likely due to the infrequency of such phrases in the pre-training data.

Table 2 compares soft and hard prefixes. For sentiment control, hard prefixes perform better, probably due to the use of "*Very*" and the model's strong ability to follow instruction on this task. In contrast, for topic control and detoxification, soft prefixes outperform hard ones, possibly due to the model's limited exposure to these attribute types during pre-training. In terms of fluency and diversity, both types show comparable performance. ***These results motivate more task-specific and flexible selection of the optimal prefix type.***



(a) (b)

Figure 1: (a) Attribute Control performance of Air-Decoding on three CTG tasks across different maximum generation lengths. Acc denotes accuracy (the higher, the better), and Tox. denotes average toxicity (the lower, the better). (b) Attention to attribute prefixes vs. generated sequence length on topic control.

Length	First Half	Second Half	Full
128	**96.58**	91.58	91.78
256	**93.18**	85.80	86.95

Table 3: Topic relevance of different segments of the outputs.

3.2 The Performance in Long Text Generation.

In Air-Decoding, the maximum generation length is fixed to 50, which is insufficient for many real-world CTG scenarios. To examine its performance in long-form generation, we vary the maximum length and evaluate attribute controllability on the three tasks. As shown in Figure 1(a), attribute controllability consistently declines with increasing sequence length. Furthermore, we divide the outputs under lengths of 128 and 256 on topic control into two halves and assess their topic relevance. As shown in Table 3, the second half exhibits significantly lower relevance, which also constrains the overall attribute relevance. This confirms the degradation in attribute control as generation progresses.

Air-Decoding models an attribute classifier via prefix-conditional language models to modulate output distribution. Inspired by the findings in multimodal domain that image attention decay leads to more hallucinations during generation (Tang et al. 2025; Yang et al. 2025), we visualize the evolution of prefix attention across the generation process. As shown in Figure 1(b) under a maximum generation length of 128 for topic control (others in Appendix C), prefix attention declines with length at a diminishing rate. This trend may arise from the nature of autoregressive generation, where each new token dilutes attention across previous tokens. A coarse theoretical analysis is in Appendix D. ***Based on this observation, we hypothesize that reduced attention to prefixes weakens the attribute classifier's reliability and thus diminishes control, which motivates us to strengthen the model's focus on the prefixes during generation.***

4 Methodology

Building on the above insights, we propose DTPA, a lightweight and efficient CTG framework based on Air-Decoding. It begins by selecting the optimal prefix type for a given task, then dynamically enhances attention to the prefix for the attribute distribution. Optionally, a similar augmentation is applied to the original prompt for the raw distribution to maintain fluency. Last we reconstruct the attribute distributions. The overall framework is illustrated in Figure 2.

4.1 Preliminary

For clarity, we illustrate our method using the sentiment control task, which includes positive and negative attributes, while it also generalizes to other CTG tasks. Given a prompt $x_{1:T-1} = \{x_1, x_2, \cdots, x_{T-1}\}$ (e.g., "*The child*" in Figure 2) and a target attribute a (e.g., positive), the goal of CTG is to guide an autoregressive model $\mathcal{G}$ (e.g., GPT-2) to generate a continuation $x_{T:N} = \{x_T, x_{T+1}, \cdots, x_N\}$ aligned with attribute a. This process is formulated as:

$$P(x_{T:N}|x_{1:T-1}, a) = \Pi_{t=T}^{N} P(x_t|x_{<t}, a). \quad (1)$$

To compute $P(x_{T:N}|x_{1:T-1})$, it is necessary to model the conditional generation probability $P(x_t|x_{<t}, a)$, which can be reformulated using Bayesian factorization (Yang and Klein 2021) (see Appendix E for more details):

$$P(x_t|x_{<t}, a) \propto P(a|x_{1:t})^{\omega} P(x_t|x_{<t}), \quad t \geq T \quad (2)$$

where $P(x_t|x_{<t})$ is the base model's next token distribution, and $P(a|x_{1:t})$ is a binary classifier $\mathcal{B}$ indicating how likely the text $x_{1:t}$ expresses attribute a. Since $x_{1:t-1}$ is fixed when generating x_t, $P(a|x_{1:t})$ reflects the relative contribution of each candidate token x_t in the vocabulary to attribute a, which we refer to as **Attribute Weight Distribution**. The parameter ω controls the strength of attribute enforcement.

Following Air-Decoding, $P(a|x_{1:t})$ is computed using two class-conditional models $P_{\phi_a}(x_t|x_{<t}, a)$ and $P_{\phi_{\bar{a}}}(x_t|x_{<t}, \bar{a})$ via Bayes' rule (Krause et al. 2021):

$$P(a|x_{1:t}) = \frac{P(a)\Pi_{j=T}^{t} P_{\phi_a}(x_j|x_{<j}, a)}{\sum_{a' \in a, \bar{a}} \Pi_{j=T}^{t} P(a') P_{\phi_{a'}}(x_j|x_{<j}, a')}, \quad (3)$$

where a and $\bar{a}$ denote positive and negative attributes, and $\phi_a, \phi_{\bar{a}}$ are the parameters of corresponding class-conditional models. When generating x_t, we only compute $P_{\phi_a}(x_t \mid x_{<t}, a)$ and $P_{\phi_{\bar{a}}}(x_t \mid x_{<t}, \bar{a})$, as all previous tokens x_j for $T \leq j < t$ have already been sampled and their corresponding probability terms $P_{\phi_{a'}}(x_j \mid x_{<j}, a')$ are thus constants and can be reused during decoding.

4.2 Attribute Prefix Augmentation

Following Air-Decoding, we model the class-conditional distributions in Eq.3 using models incorporating attribute prefixes. For soft prefixes, the learnable parameters $\theta_{a'}$ are optimized on attribute-specific datasets by minimizing:

$$\mathcal{L}_{LM} = -\sum_{k=1}^{K} log P_{\lambda, \theta_{a'}}(x_k|x_{<k}, H_{\theta_{a'}}), \quad (4)$$

where λ denotes frozen GPT-2 parameters, and $a^{'} \in \{a, \bar{a}\}$ is the attribute. For hard prefixes, no training is required, and $\theta_{a'} \in \lambda$. Combining these prefixes with GPT-2 yields Prefix-Conditional Language Models (PC-LMs), which are used to represent $P_{\phi_a}(x_t|x_{<t}, a)$ and $P_{\phi_{\bar{a}}}(x_t|x_{<t}, \bar{a})$ in Eq.3 as $P_{\lambda,\theta_a}(x_t|x_{<t}, H_{\theta_a})$ and $P_{\lambda,\theta_{\bar{a}}}(x_t|x_{<t}, H_{\theta_{\bar{a}}})$, denoted as **attribute distributions**. Then Eq.3 is reformulated as:

$$P(a|x_{1:t}) = \frac{\Pi_{j=T}^{t} P_{\lambda,\theta_a}(x_j|x_{<j}, H_{\theta_a})}{\sum_{a' \in a, \bar{a}} \Pi_{j=T}^{t} P_{\lambda,\theta_{a'}}(x_j|x_{<j}, H_{\theta_{a'}})}, \quad (5)$$

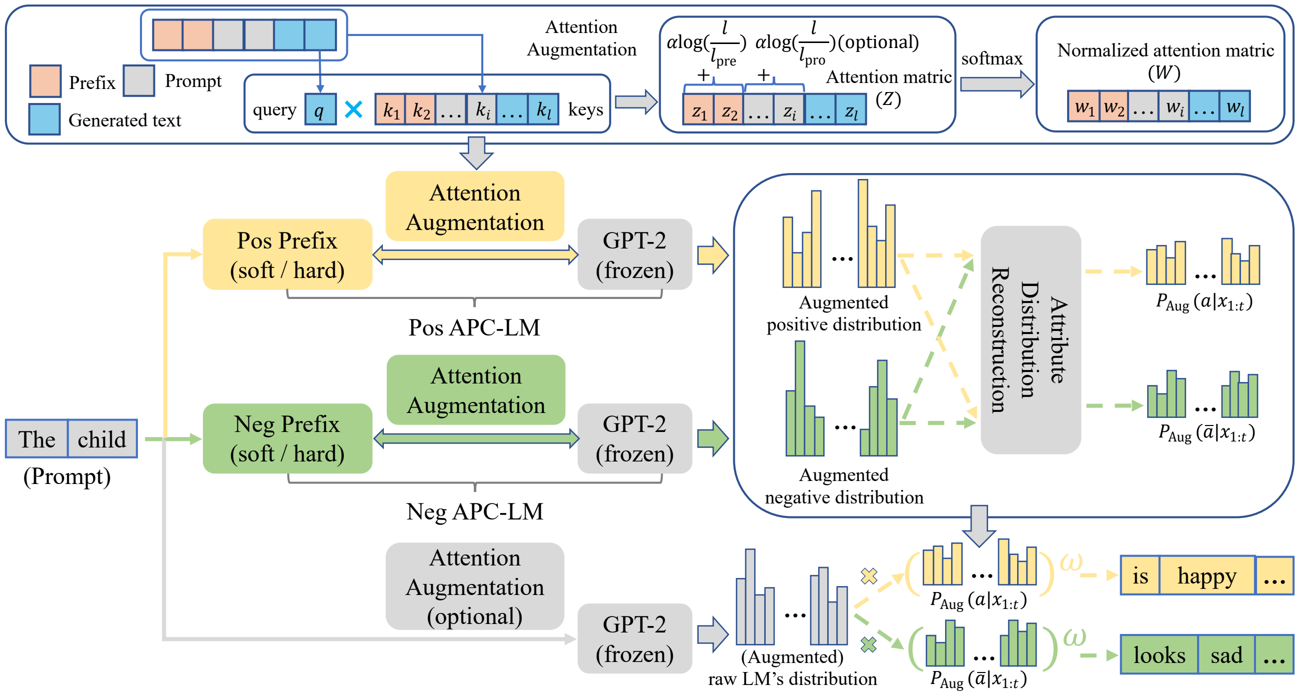


Figure 2: Overall framework of DTPA. Given a prompt "*The child*", the Pos APC-LM and Neg APC-LM generate the augmented positive and negative distribution, respectively. After attribute distribution reconstruction, we obtain the augmented attribute weight distributions $P_{\text{Aug}}(a|x_{1:t})$ and $P_{\text{Aug}}(\bar{a}|x_{1:t})$. Based on the task, we optionally apply a similar augmentation to original prompt for the raw LM to get the (augmented) raw distribution. Finally, the (augmented) raw distribution is weighted by the corresponding augmented attribute weight distribution to generate next token.

Considering the equal training data for soft prefixes and the assumed equal class priors in GPT-2 for hard prefixes, $P(a)$ and $P(\bar{a})$ can be omitted.

As discussed in Section 3.2, prefix attention decay potentially weakens the attribute classifier and thus impair controllability. To mitigate this, we enhance prefix attention dynamically. Specifically, for the last token in the current sequence, we apply a sequence-length-dependent scaling factor to its attention scores over the prefix region in the pre-softmax attention matrix. Notably, here we employ KV Cache to accelerate generation, where the attention scores are computed via the dot product between the query of the last token and all tokens' keys, resulting in an attention matrix $Z \in \mathbb{R}^{1\times l}$, with $l = l_{\text{pre}} + l_{\text{pro}} + l_{\text{gen}}$ (where l_{pre}, l_{pro}, and l_{gen} denote the lengths of the prefix, prompt, and generated text, respectively). For simplicity, we uniformly denote the attention matrix in each layer and head as $Z = [z_1, z_2, ..., z_l]$. The attention matrix after scaling is:

$$z_i = \begin{cases} qk_i + \alpha \log(\frac{l}{l_{\text{pre}}}), & 1 \le i \le l_{\text{pre}} \\ qk_i, & l_{\text{pre}} < i \le l \end{cases} \tag{6}$$

The softmax-normalized attention matrix is:

$$w_i = \text{softmax}(z_i) = \begin{cases} \dfrac{(\frac{l}{l_{\text{pre}}})^{\alpha} e^{qk_i}}{\sum_{j=1}^{l_{\text{pre}}} (\frac{l}{l_{\text{pre}}})^{\alpha} e^{qk_j} + \sum_{j=l_{\text{pre}}+1}^{l} e^{qk_j}}, & 1 \le i \le l_{\text{pre}} \\ \dfrac{e^{qk_i}}{\sum_{j=1}^{l_{\text{pre}}} (\frac{l}{l_{\text{pre}}})^{\alpha} e^{qk_j} + \sum_{j=l_{\text{pre}}+1}^{l} e^{qk_j}}, & l_{\text{pre}} < i \le l \end{cases} \tag{7}$$

where q is the query of the last token and k_i is the key of the i-th token. α is a tunable exponent to modulate the strength of prefix attention amplification. When $\alpha = 1$, the attention to prefix tokens approximately scales linearly with l, assuming the denominator remains stable. This method dynamically augments the prefix attention at each generation step, hence we term it Dynamic Token-level Prefix Augmentation (DTPA). The intuition behind it is presented in Appendix F.

This process yields Augmented Prefix-Conditional Models (APC-LMs), generating **augmented attribute distributions**, as shown in Figure 2. Accordingly, Eq.5 becomes:

$$P_{\text{Aug}}(a|x_{1:t}) = \frac{\Pi_{j=T}^{t} P_{\lambda,\theta_a}(x_j|x_{<j}, \text{Aug}(H_{\theta_a}))}{\sum_{a' \in a,\bar{a}} \Pi_{j=T}^{t} P_{\lambda,\theta_{a'}}(x_j|x_{<j}, \text{Aug}(H_{\theta_{a'}}))}, \tag{8}$$

where $\text{Aug}(H_{\theta_{a'}})$ denotes augmenting the prefix.

4.3 Original Prompt Augmentation

Another important concern is that the original prompt (e.g., "*The child*" in Figure 2) also suffers from attention dilution as generation progresses. Enhancing prefix attention may further diminish the model's focus on the prompt, weakening semantic alignment and potentially causing the output distribution to drift from the intended one. To mitigate this, we apply a similar attention augmentation to the prompt for the raw distribution rather than the attribute distribution, to avoid interference with attribute prefix augmentation. The corresponding modified attention matrix is given by:

$$w_i = \begin{cases} \dfrac{(\frac{l}{l_{\text{pro}}})^{\alpha} e^{qk_i}}{\sum_{j=1}^{l_{\text{pro}}}(\frac{l}{l_{\text{pro}}})^{\alpha} e^{qk_j} + \sum_{j=l_{\text{pro}}+1}^{l} e^{qk_j}}, & 1 \le i \le l_{\text{pro}} \\ \dfrac{e^{qk_i}}{\sum_{j=1}^{l_{\text{pro}}}(\frac{l}{l_{\text{pro}}})^{\alpha} e^{qk_j} + \sum_{j=l_{\text{pro}}+1}^{l} e^{qk_j}}, & l_{\text{pro}} < i \le l \end{cases} \tag{9}$$

where $l = l_{\text{pro}} + l_{\text{gen}}$ since prefixes are not used in obtaining the raw distribution, and α is consistent with that used for prefix augmentation to avoid additional hyperparameter tuning. The resulting augmented raw distribution is:

$$P_{\text{Aug}}(x_t|x_{<t}) = P(x_t|x_{<t}, \text{Aug}(\text{Prompt})). \tag{10}$$

In practice, we find that prompt augmentation benefits sentiment and topic control but harms detoxification (see Appendix J.1). We attribute this to the use of challenging prompts in detoxification, which are prone to eliciting toxic content. Augmenting such prompts may worsen generation. Thus, we remove this component specifically for this task.

4.4 Attribute Distribution Reconstruction

With prefix augmentation, APC-LM may exhibit a strong tendency toward the desired attribute, allocating significantly higher weights for attribute-related words compared to non-attribute-related words. This overemphasis may cause the model to disproportionately favor attribute words when predicting the next token, potentially leading to the Attribute Collapse phenomenon. To mitigate it, we follow Air-Decoding and reconstruct the augmented attribute distributions by applying an inverse-log transformation to them, as formulated below:

$$\widetilde{P}_{\lambda,\theta_a}(x_j|x_{<j}, \text{Aug}(H_{\theta_a})) = -\frac{1}{\log\left(P_{\lambda,\theta_a}(x_j|x_{<j}, \text{Aug}(H_{\theta_a}))\right)} \tag{11}$$

$$\widetilde{P}_{\lambda,\theta_{\bar{a}}}(x_j|x_{<j}, \text{Aug}(H_{\theta_{\bar{a}}})) = -\frac{1}{\log\left(P_{\lambda,\theta_{\bar{a}}}(x_j|x_{<j}, \text{Aug}(H_{\theta_{\bar{a}}}))\right)} \tag{12}$$

$$P_{\text{Aug}}(a|x_{1:t}) = \frac{\Pi_{j=T}^{t}\widetilde{P}_{\lambda,\theta_a}(x_j|x_{<j}, \text{Aug}(H_{\theta_a}))}{\sum_{a' \in \{a,\bar{a}\}}\Pi_{j=T}^{t}\widetilde{P}_{\lambda,\theta_{a'}}(x_j|x_{<j}, \text{Aug}(H_{\theta_{a'}}))} \tag{13}$$

Through the above reconstruction, the augmented attribute distributions are compressed into a smaller range interval without altering the relative order of their elements, thereby achieving a more balanced weight distribution between attribute-related and non-attribute-related words. For better understanding, we provide an illustrative example in Appendix G.

5 Experiments

5.1 Evaluation Metric

Following Air-Decoding, we conduct both automatic and human evaluation to assess the quality of generated texts.

Automatic Evaluation. We mainly focus on three aspects of generated texts. (1) **Accuracy**: For sentiment control and topic control, we adopt the RoBERTa (Liu et al. 2019) classifiers trained by Air-Decoding on Yelp Review and AG-News dataset (Zhang, Zhao, and LeCun 2015) to calculate attribute accuracy (Acc), which achieve 98.53% and 95.57% on their corresponding test sets. For detoxification, we utilize the Perspective API[1] to measure average toxicity. (2) **Fluency**: Evaluated by perplexity (PPL) calculated by GPT-2 large; lower PPL indicates higher fluency. (3) **Diversity**: We compute the distinctness (Li et al. 2016) of 1-grams, 2-grams, and 3-grams (Dist-1, Dist-2, and Dist-3) for generated texts to assess the diversity.

Human Evaluation. We follow Zhang and Song (2022) and assess generated texts from three aspects. (1) **Relevance (Rel.)** measures how well the generated text satisfies the desired attribute. (2) **Fluency (Flu.)** judges the fluency of the generated text from a human perspective. (3) **Topicality (Top.)** evaluates the consistency between the generated text and the given prompt. For each task, 100 random samples are rated by three annotators on a scale from 1 (very poor) to 5 (very good) (see more details in Appendix H). Final scores are averaged over 300 ratings per metric.

5.2 Baselines

We compare DTPA with a wide range of baselines: (1) **Prefix-Tuning** controls text generation by training prefixes while keeping the language model frozen. (2) **Contrastive Prefixes (Con Prefixes)** achieve controllable generation by jointly training contrastive prefixes. (3) **Discup** uses an attribute discriminator to optimize prompts for guiding generation. (4) **PPLM** modifies hidden states via gradients from an attribute classifier. (5) **GeDi** (Krause et al. 2021) employs a class-conditional LM to adjust the output distribution. (6) **DExperts** fine-tunes expert/anti-expert models and performs contrastive decoding. (7) **LM-Steer** (Han et al. 2024) uses word embeddings to steer language model generation styles. (8) **FreeCtrl** identifies attribute-relevant FFN vectors and adaptively adjusts their weights. (9) **Air-Decoding** reconstructs attribute distributions from PC-LMs to balance weights between attribute and non-attribute words.

For LM-Steer and FreeCtrl, we implement them using their public codes. For other baselines, we use the results from Air-Decoding for consistency and fairness. All methods (except Discup[2]) use GPT-2 medium as the base model, and follow Air-Decoding's decoding strategy: top-k sampling with $k = 200$ (Discup uses $k = 20$). More hyperparameter details can be found in Appendix I.

5.3 Experimental Setup

Sentiment Control. We use the hard prefixes noted in section 3.1. The 15 prompts used in PPLM are adopt for evaluation. A total of 100 sentences (50 positive and 50 negative) with a length of 50 are generated for each of the 15 prompts. In our method, the hyperparameter ω is is set to 140.0, consistent with Air-Decoding, and α is set to $1/2$.

Topic Control. We use the soft prefixes trained by Air-Decoding, each with a length of 20. The 20 prompts used for evaluation are from PPLM. For each prompt, we generate 50 sentences for each attribute from **World**, **Sports**, **Business**, and **Science**, each with a length of 50. ω is set to 60.0 as in Air-Decoding and α is set to $1/2$.

[1]https://www.perspectiveapi.com/

[2]Discup only released prompts for GPT-2 large.

Method	Acc	PPL↓	Dist-1	Dist-2	Dist-3
Pre-Tuning	62.15	38.49	0.11	0.53	0.82
Con Prefixes	75.66	35.32	0.11	0.52	0.81
Discup*	95.20	39.14	0.07	0.46	0.80
PPLM	69.06	34.89	0.12	0.51	0.77
GeDi	94.23	169.86	0.15	0.53	0.74
DExpert	94.74	51.99	**0.16**	**0.65**	**0.85**
LM-Steer	90.20	61.49	0.07	0.35	0.59
FreeCtrl	95.07	146.83	0.08	0.51	0.82
Air-Decoding	96.82	**26.66**	0.13	0.55	0.78
DTPA	**98.07**	38.03	0.13	0.57	0.80

Table 4: The main experimental results on sentiment control. ↓ suggests that the performance is better with a lower score. $*$ means the backbone model is GPT-2 large.

Method	Acc	PPL↓	Dist-1	Dist-2	Dist-3
Pre-Tuning	72.74	64.43	0.09	0.49	0.74
Con Prefixes	88.47	70.34	0.09	**0.50**	**0.75**
GeDi	94.27	104.46	**0.10**	0.48	0.69
FreeCtrl	80.03	103.93	0.07	0.52	0.83
Air-Decoding	97.21	**31.18**	0.08	0.47	0.74
DTPA	**98.15**	39.42	0.08	0.47	0.74

Table 5: The main experimental results on topic control. As LM-Steer hasn't evaluated on this task, we don't include it.

Detoxification. We adopt the soft prefixes trained by Air-Decoding, each with a length of 20. The prompts are the same as in Air-Decoding, sourced from RealToxicityPrompts (Gehman et al. 2020) and categorized as "challenging" with toxicity scores no larger than 0.5, evaluated by Perspective. We generate 20 sentences per prompt for all 203 selected prompts, each with a length of 50. ω is set to 120.0 as in Air-Decoding and α is set to $1/3$.

All experiments are conducted on a single NVIDIA 4090 GPU for a fair comparison.

5.4 Main Results and Analysis

Sentiment Control. As shown in Table 4, DTPA achieves the best attribute control performance on sentiment control in automatic evaluations, outperforming Air-Decoding by 1.29%, which is non-trivial given the already high baseline. This gain can be attributed to the meticulously designed hard prefixes and dynamic token-level prefix augmentation. Notably, on this task, DTPA requires no training due to the use of hard prefixes and surpasses the recent powerful training-free approach Free-Ctrl by 3 points, highlighting its strong effectiveness. While DTPA improves attribute relevance, it slightly compromises fluency. This may stem from the reduced contextual focus caused by stronger prefix attention. In terms of diversity, DTPA surpasses Air-Decoding on Dist-2 and Dist-3, indicating a higher text diversity.

Topic Control. For topic control, we show the results in Table 5, where DTPA similarly achieves SOTA performance in attribute relevance. At the same time, it maintains competitive fluency, ranking just behind Air-Decoding, and performs comparably in diversity. These results indicate that DTPA remains effective in multi-class attribute control by enhancing the attention to the attribute prefix and original prompt, thereby improving control performance while maintaining overall text quality.

Method	Tox.↓	PPL↓	Dist-1	Dist-2	Dist-3
Pre-Tuning	49.2	92.20	0.07	0.40	0.68
Con Prefixes	21.7	85.34	-	-	-
Discup*	**14.8**	63.90	0.07	0.48	0.82
PPLM	30.0	148.50	-	-	-
GeDi	20.5	166.01	-	-	-
DExpert	20.0	58.06	0.08	0.48	0.78
LM-Steer	39.4	83.52	0.06	0.39	0.66
Air-Decoding	18.5	**48.29**	0.07	0.44	0.74
DTPA	18.4	48.34	0.07	0.44	0.74

Table 6: The main experimental results on detoxification. Since FreeCtrl didn't provide the attribute keywords for this task which are important for reproduction, it's not included.

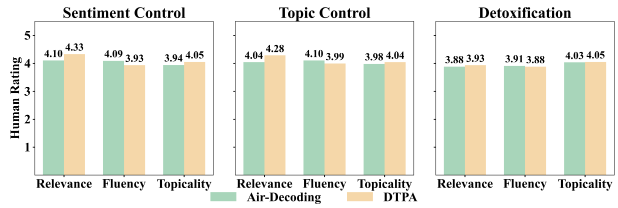


Figure 3: Comparison of human evaluation results for DTPA and Air-Decoding.

Detoxification. As shown in Table 6, DTPA achieves an average toxicity score of 18.4, second only to Discup. Its limited advantage over Air-Decoding in toxicity control may stem from the weak semantic representations of the trained prefixes, leading to a less effective attribute classifier and a blurred distinction between toxic and nontoxic tokens. In terms of fluency and diversity, DTPA performs comparably to Air-Decoding, both significantly surpassing other baselines, confirming the effectiveness of attribute distribution reconstruction in balancing controllability and text quality.

Human Evaluations. We also conduct human evaluations on the three tasks to compare the texts generated by DTPA and Air-Decoding, as illustrated in Figure 3. The human evaluation results are consistent with automatic evaluations, where DTPA demonstrates stronger attribute controllability but suffers a slight decline in fluency. In terms of topicality, our method performs comparably to Air-Decoding. These results further confirm the effectiveness of our method.

5.5 Ablation Studies

In this section, we conduct ablation studies on sentiment control to assess the effects of different factors. The results of another two tasks are provided in Appendix J.

The Effects of Different Components. We conduct ablation studies to analyze the effects of DTPA's different com-

#	Pre-Aug	Pro-Aug	ADR	Acc	PPL↓	Dist-1	Dist-2	Dist-3
1	✗	✗	✗	95.93	38.70	**0.15**	**0.62**	**0.83**
2	✗	✗	✓	97.40	29.69	0.14	0.57	0.79
3	✓	✗	✓	97.80	**29.59**	0.14	0.57	0.79
4	✗	✓	✓	97.27	38.75	0.14	0.58	0.80
5	✓	✓	✓	**98.07**	38.03	0.14	0.57	0.80

Table 7: The effects of different components on sentiment control.

α	Acc	PPL↓	Dist-1	Dist-2	Dist-3
1	97.40	71.43	0.13	0.55	0.78
$\mathbf{1/2}$	**98.07**	38.03	0.13	0.57	**0.80**
$1/3$	97.07	33.75	**0.14**	**0.58**	0.80
$1/4$	97.27	**32.30**	0.14	0.58	0.79

Table 8: Performance of DTPA under different attention scaling exponents α on sentiment control.

ponents, including Prefix Augmentation (Pre-Aug), Prompt Augmentation (Pro-Aug), and Attribute Distribution Reconstruction (ADR), as shown in Table 7. ADR notably enhances attribute control and fluency by balancing the weights of attribute and non-attribute words. Augmenting only the prefix or prompt yields limited gains: prefix-only augmentation may overemphasize attribute tokens and shift attention away from context, while prompt-only augmentation reduces the probability of selecting attribute-relevant tokens. In contrast, combining both achieves a better trade-off between attribute focus and contextual coherence.

The impact of Attention Scaling Exponent α. The exponent α controls the extent of attention enhancement for the prefix and prompt. We evaluate several representative values, with results presented in Table 8. We find that either overly large or small values yield suboptimal performance. A large α leads to excessive focus on the attribute prefix, distorting the output distribution and reducing text fluency. Conversely, a small α results in limited enhancement, making DTPA behave similarly to Air-Decoding, with improved fluency but diminished control. We set α based on the results in our main experiments. These values serve as a baseline of our method rather than theoretically optimal choices.

5.6 Further Analysis

Performance of DTPA in Long Text Generation. We compare DTPA and Air-Decoding under four maximum generation lengths (64, 128, 192, 256) across the three CTG tasks. As shown in Table 9 for sentiment control (others in Appendix K), DTPA consistently surpasses Air-Decoding in attribute relevance, with the gap widening at longer lengths. DTPA maintains around 97% accuracy even at length 256, while Air-Decoding drops to 92.53%, demonstrating the efficacy of prefix attention augmentation. While this gain comes with a slight fluency drop, it remains acceptable. DTPA also exhibits marginal improvements in diversity. Overall, by dynamically enhancing prefix and prompt attention, DTPA delivers stronger attribute control with competitive text quality in long text generation.

Length	Method	Acc	PPL↓	Dist-1	Dist-2	Dist-3
64	Air-Decoding	95.13	**25.64**	0.12	0.55	0.80
	DTPA	**97.93**	36.69	**0.12**	**0.58**	**0.82**
128	Air-Decoding	96.53	**23.93**	0.09	0.52	0.82
	DTPA	**97.33**	34.26	**0.09**	**0.54**	**0.84**
192	Air-Decoding	94.27	**23.09**	0.09	0.50	0.82
	DTPA	**96.87**	33.86	**0.07**	**0.50**	**0.83**
256	Air-Decoding	92.53	**23.35**	0.06	0.47	0.81
	DTPA	**97.40**	34.06	**0.06**	**0.48**	**0.82**

Table 9: Performance comparison between DTPA and Air-Decoding under different generation lengths on sentiment control.

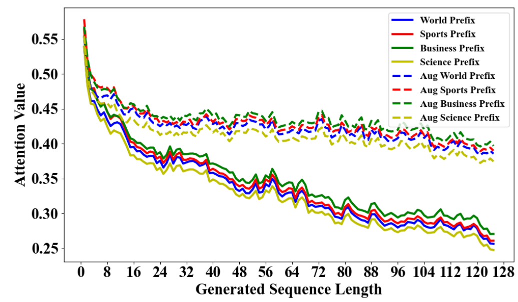


Figure 4: Visualization of prefix attention on topic control task before and after augmentation.

Visualization of Prefix Attention Augmentation. To better illustrate the effect of DTPA on prefix attention, we visualize the prefix attention values before and after augmentation at a maximum generation length of 128. As shown in Figure 4 for the topic control task (additional results in Appendix L), prefix attention is notably increased after augmentation, with a slower decay as the sequence grows. This demonstrates that DTPA effectively mitigates attention dilution over prefixes, thereby improving attribute focus and enhancing controllability.

Case Studies. We provide case studies in Appendix M to further illustrate the effectiveness of DTPA in CTG.

6 Conclusion

In this paper, we conduct an in-depth investigation into the impact of different prefix types and the decline of attribute controllability in long text generation based on Air-Decoding, which we hypothesize is primarily caused by the attention decay to the prefixes over longer sequences. Based on these insights, we propose DTPA, a lightweight and effective CTG framework that selects the optimal prefix type and dynamically enhances prefix attention with an exponentially increasing scaling factor as sequence length grows. Prompt augmentation is optionally applied based on the task. Experiments on multiple CTG tasks demonstrate that DTPA achieves superior controllability while preserving overall text quality, particularly in long text generation.

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A Limitations and Future Work

Despite DTPA demonstrates significant advantages across multiple CTG tasks, especially in long text generation, it still presents several limitations. On the one hand, the improvement brought by DTPA on detoxification is relatively limited. On the other hand, the fluency of generated texts still leaves room for improvement. To further push the performance ceiling of DTPA, we will explore the design and integration of more powerful prefixes. Moreover, extending DTPA to other prefix-based CTG methods represents a promising research direction. Applying DTPA to large-scale language models such as LLaMA-2, as well as exploring its capability in multi-attribute control, are also meaningful avenues for future work.

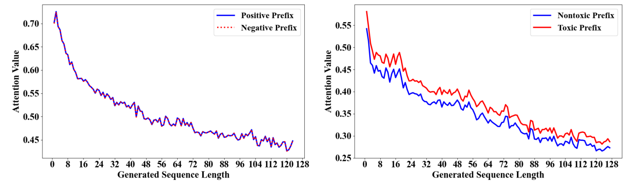
B Comparison of Different Hard Prefixes

This section presents a comparison among various formats of hard prefixes to underscore the effectiveness of our meticulously designed hard prefixes introduced in Section 3.1. The results are shown in Table 10. We design multiple formats of attribute prefixes for each task, but only present one representative prefix per format to simplify the presentation. Prefixes for other attributes follow the same format with only the attribute word modified. For example, in the sentiment control task, the prefix for the positive attribute in the first format is "*A positive text:*" (as shown in the table), then the hard prefix for the negative attribute in the same format would be "*A negative text:*", and so on. From the table, we can observe the following interesting patterns and insights.

For sentiment control and detoxification, where both attributes are described using adjectives, we find that adding the degree adverb "*Very*" significantly enhances the effectiveness of attribute control. This may be attributed to the inherent instruction-following ability of GPT-2. In contrast, for topic control, the opposite trend is observed: the prefix "*World-related*" outperforms its counterpart with "*Very*". A possible explanation is that GPT-2 has been exposed to a greater number of "*World-related*" expressions during pretraining, whereas phrases like "*Very world-related*" are less frequent or even unnatural. As a result, the model may struggle to accurately interpret and respond to the latter, leading to less effective topic control. Meanwhile, we also find that shorter hard prefixes tend to yield stronger effects. This can be explained by the fact that shorter instructions help the model focus more on the attribute words, thereby strengthening the control performance. Such a phenomenon is consistently observed across all three tasks.

Overall, the best-performing hard prefixes selected for each task share two key characteristics: they are concise and convey clear attribute guidance. This observation offers useful insights into designing hard prefixes for other attribute control tasks.

Nevertheless, considering the vast combinatorial space of words and phrases in natural language, it is infeasible to exhaustively explore all possible prefix forms. The selected hard prefixes only serve as a baseline for our method and are expected to provide inspiration for future research on prompt engineering and attribute control.



(a) Sentiment Control (b) Detoxification

Figure 5: Attention to attribute prefixes vs. generated sequence length on (a) sentiment control and (b) detoxification.

C Visualization of Attribute Prefix Attention

In Section 3.2, we visualize how the attention to attribute prefixes changes as sequence length increases on topic control. Here we present the visualization results for sentiment control and detoxification in Figure 5(a) and 5(b), respectively. It can be observed that the attention to attribute prefixes exhibits a consistent declining trend as the sequence grows across different CTG tasks. This declining attention trend aligns with the observed drop in attribute control performance across tasks, highlighting a potentially strong link between them. Notably, the prefixes used in sentiment control are hard prefixes, yet still exhibit an attention degradation trend, suggesting that this phenomenon is insensitive to the prefix type.

D A Coarse Theoretical Analysis of Prefix Attention Degradation

Based on the following special case, we provide a coarse theoretical analysis to explain the phenomenon of prefix attention degradation observed in Section 3.2. Suppose that during text generation, the last token in the sequence attends equally to all previous tokens, including the attribute prefix. Let the lengths of the attribute prefix, original prompt and generated text be l_{pre}, l_{pro} and l_{gen}, respectively. Then, the attention allocated to the prefix in each transformer layer and each attention head can be computed as:

$$\text{Attention}_{\text{prefix}} = \frac{l_{\text{pre}}}{l_{\text{pre}} + l_{\text{pro}} + l_{\text{gen}}} \tag{14}$$

Since l_{pre} and l_{pro} are fixed, the attention to the prefix decreases as the generated sequence length l_{gen} increases, following an inverse proportional relationship, approximately consistent with the curves shown in Figure 1(b), Figure 5(a), and Figure 5(b).

E Bayesian Factorization

Following Zhong et al. (2023), we present the derivation of Eq.2 and Eq.3 in Section 4.1 based on Bayesian Factorization as follows:

Task	Hard Prefix	Acc / Tox.	PPL↓	Dist-1	Dist-2	Dist-3
Sentiment Control	`"A positive text:"`	89.60	28.48	0.13	0.56	0.78
	`"A very positive text:"`	97.20	28.37	0.13	0.55	0.78
	`"Positive:"`	91.27	**26.75**	0.13	0.55	0.76
	`"Very positive:"`	**97.40**	29.69	**0.14**	**0.57**	**0.79**
Topic Control	`"A text about world"`	63.53	34.86	0.07	0.44	0.72
	`"About world:"`	70.68	31.03	0.06	0.42	0.71
	`"The topic is world:"`	70.15	31.11	0.07	0.43	0.72
	`"World topic:"`	73.85	34.59	0.08	0.46	0.74
	`"World-related:"`	**80.30**	31.58	0.07	**0.46**	0.73
	`"Very world-related:"`	74.50	**28.78**	**0.08**	0.46	**0.74**
Detoxification	`"A nontoxic text:"`	29.7	50.95	0.07	0.45	0.75
	`"A very nontoxic text:"`	29.7	**49.84**	**0.08**	**0.47**	**0.77**
	`"Nontoxic:"`	36.3	69.47	0.07	0.46	0.76
	`"Very nontoxic:"`	**26.3**	50.45	0.07	0.46	0.76

Table 10: Performance comparison of Air-Decoding using different hard prefixes.

$$
\begin{aligned}
P(x_t|x_{1:t-1}, a) &= \frac{P(x_{1:t}, a)}{P(x_{1:t-1}, a)} \\
&= \frac{P(a|x_{1:t})P(x_{1:t})}{P(x_{1:t-1}, a)} \\
&= \frac{P(a|x_{1:t})P(x_t|x_{1:t-1})P(x_{1:t-1})}{P(a|x_{1:t-1})P(x_{1:t-1})} \\
&= \frac{P(a|x_{1:t})P(x_t|x_{1:t-1})}{P(a|x_{1:t-1})} \\
&\propto P(x|x_{1:t})P(x_t|x_{1:t-1})
\end{aligned}
$$

$$
\begin{aligned}
P(a|x_{1:t}) &= \frac{P(x_{1,t}, a)}{P(x_{1:t})} \\
&= \frac{P(a)P_{\phi_a}(x_{1:t}|a)}{\sum_{a' \in \{a,\bar{a}\}} P(a')P_{\phi_{a'}}(x_{1:t}|a')} \\
&= \frac{P(a)\Pi_{j=T}^{t} P_{\phi_a}(x_j|x_{<j}, a)}{\sum_{a' \in \{a,\bar{a}\}} \Pi_{j=T}^{t} P(a')P_{\phi_{a'}}(x_j|x_{<j}, a')}
\end{aligned}
$$

F The Intuition Behind Our Prefix Attention Augmentation Mechanism

In Appendix D, we use a specific example to reveal the underlying cause of attribute prefix degradation. Building on this, we now delve into the intuition behind our attention augmentation mechanism. From Eq.14, we observe that the attention to attribute prefixes is approximately inversely proportional to the sequence length l ($l = l_{\text{pre}} + l_{\text{pro}} + l_{\text{gen}}$). To alleviate this degradation, a natural and straightforward idea is to multiply the normalized attention scores of attribute prefixes by a coefficient dependent on the sequence length. However, this approach breaks the probabilistic constraint that attention scores must sum to 1. To avoid this issue, we instead add a logarithmic length-dependent scaling factor (e.g., $\log(l/l_{\text{pre}})$) to the unnormalized attention logits before the softmax operation. This modification is approximately equivalent to multiplying the normalized attention scores by a length-dependent factor (e.g., l/l_{pre}) after softmax normalization (assuming the change in the softmax denominator is negligible), while still satisfying the probabilistic constraint that the attention scores sum to 1.

Scaling Factor	Acc	PPL↓	Dist-1	Dist-2	Dist-3
$\alpha \log(l/(l_{\text{pre}} + l_{\text{pro}}))$	97.47	38.58	**0.14**	**0.58**	0.80
$\alpha \log(l/l_{\text{pre}})$	**98.07**	**38.03**	0.13	0.57	**0.80**

Table 11: Comparison results between two scaling factors on sentiment control.

Specifically, when generating the first token (i.e., $l_{\text{gen}} = 0$), the prefix attention is $l_{\text{pre}}/(l_{\text{pre}} + l_{\text{pro}})$. To maintain a consistent level of prefix attention throughout the generation process, we consider using a scaling factor of $\log(l/(l_{\text{pre}} + l_{\text{pro}}))$. This ensures that at each generation step, the prefix attention approaches $l_{\text{pre}}/l \cdot l/(l_{\text{pre}} + l_{\text{pro}}) = l_{\text{pre}}/(l_{\text{pre}} + l_{\text{pro}})$, thus maintaining stability during generation.

However, a linear scaling with respect to l may significantly suppress the attention to non-prefix tokens, which could negatively affect model performance. To strike a balance, we introduce a tunable coefficient α. After softmax operation, α effectively acts as an exponent, so we consider common fractional values such as $1, 1/2, 1/3$, etc.

Alternatively, setting the denominator of the scaling factor to l_{pre} leads to a stronger amplification of the prefix attention. We compare two scaling factors on sentiment control: $\alpha \log(l/(l_{\text{pre}} + l_{\text{pro}}))$ and $\alpha \log(l/l_{\text{pre}})$, both with $\alpha = 1/2$. As shown in Table 11, the latter yields a better control performance, due to the smaller denominator resulting in stronger amplification. Based on this observation, we choose $\alpha \log(l/l_{\text{pre}})$ as the prefix attention scaling factor. Since both l and l_{pre} are fixed at inference time, only α needs to be tuned manually, providing flexibility while avoiding excessive hyperparameter tuning. The prompt augmentation follows a similar process, with the scaling factor set to $\alpha \log(l/l_{\text{pro}})$.

G An Illustrative Example of Attribute Distribution Reconstruction

To better clarify the function of attribute distribution reconstruction mentioned in Section 4.4, we provide a detailed and illustrative example as below.

As shown in Figure 2, when generating a sentence with positive attribute given the prompt "*The child*", suppose at the first decoding step, the word "*happy*" has a probability of 0.15 in the augmented positive attribute distribution $P_{\lambda,\theta_a}(x_t|x_{<t}, \text{Aug}(H_{\theta_a}))$ and 0.01 in the augmented negative attribute distribution $P_{\lambda,\theta_{\bar{a}}}(x_t|x_{<t}, \text{Aug}(H_{\theta_{\bar{a}}}))$, while the word "*is*" has a probability of 0.02 and 0.07, respectively. According to Eq.8, the resulting weights in the final attribute weight distribution $P_{\text{Aug}}(a|x_{1:t})$ for "*happy*" and "*is*" are $0.15/(0.15 + 0.01) = 0.938$ and $0.02/(0.02 + 0.07) = 0.222$, respectively. Thus, "*happy*" receives a significantly higher weight (approximately 4 times that of "*is*") and is more likely to be selected, despite being semantically inconsistent with the previous prompt "*The child*", leading to decreased fluency in the generated text.

By applying attribute distribution reconstruction, the values from the two attribute distributions are first compressed into a smaller range without altering the relative order of their elements, thereby balancing the weights between attribute-relevant and irrelevant words. In this case, the reconstructed augmented positive attribute distribution $\widetilde{P}_{\lambda,\theta_a}(x_j|x_{<j}, \text{Aug}(H_{\theta_a}))$ assigns weights of $-1/\log(0.15) = 0.527$ to "*happy*" and $-1/\log(0.02) = 0.256$ to "*is*"; in the reconstructed augmented negative attribute distribution $\widetilde{P}_{\lambda,\theta_{\bar{a}}}(x_j|x_{<j}, \text{Aug}(H_{\theta_{\bar{a}}}))$, "*happy*" and "*is*" receive weights of $-1/\log(0.01) = 0.217$ and $-1/\log(0.07) = 0.376$, respectively. Consequently, in the final attribute weight distribution, "*happy*" and "*is*" obtain weights of $0.527/(0.527 + 0.217) = 0.708$ and $0.256/(0.256 + 0.376) = 0.405$, respectively, where the weight of "*happy*" is less than twice that of "*is*". This reconstruction not only maintains a higher weight for "*happy*" but also reduces the gap between the two words, increasing the likelihood of "*is*" being selected. Such balance prevents the model from overly favoring attribute-related words, thereby improving the fluency of generated text.

H Human Evaluation Details

In this section we provide specific scoring guidelines for each human evaluation metric following Zhong et al. (2023).

H.1 Relevance

- 5: The generated sentences are perfectly aligned with the desired attribute.
- 4: The generated sentences are very related to the desired attribute.
- 3: The generated sentences are very related to the desired attribute.
- 2: The generated sentences have a relatively weak consistency with the desired attribute.
- 1: The generated sentences have no correlation with the desired attribute, and in some cases, they even contradict it.

H.2 Fluency

- 5: The generated sentences are grammatically correct, fluent, and easy to understand.
- 4: The generated sentences are grammatically correct, but they are slightly less smooth, yet still easily understandable.
- 3: The generated sentences have a few grammar errors, but they do not hinder understanding.
- 2: The generated sentences have a few grammar errors and are not very easy to understand.
- 1: The generated sentences have numerous grammar errors, lack coherence, and are difficult to understand.

H.3 Topicality

- 5: The generated sentences are grammatically correct, fluent, and easy to understand.
- 4: The generated sentences exhibit a relatively strong correlation with the input prompt.
- 3: The generated sentences have an average correlation with the input prompt.
- 2: The generated sentences have a relatively weak correlation with the input prompt.
- 1: The generated sentences have a poor correlation with the input prompt, resulting in incoherent sentences.

I Hyperparameter Details

For LM-Steer (Han et al. 2024), as the model checkpoints are not released, we follow its settings to train two LM-Steers for sentiment control and detoxification. And the hyperparameter $\epsilon_0 = 1e-3$, consistent with the original paper. On sentiment control, we use $5\epsilon_0$ as steer values for positive attribute and $-5\epsilon_0$ for negative attribute. On detoxification, we use $5\epsilon_0$ as steer values. For FreeCtrl (Feng et al. 2024), we follow its original settings and set $k = 30$, $\mu_\omega = 1.15$ for sentiment control and topic control, while setting λ to 1.5 and 0.3 for the two tasks, respectively. For other baselines, we follow the experiment settings of Air-Decoding (Zhong et al. 2023) and use their results directly. We refer readers to the paper for more details.

J More Results of Ablation Studies

J.1 The Effects of Different Components

We provide additional results of the ablation studies about the effects of different components for topic control and detoxification, which are exhibited in Table 12 and Table 13, respectively. For topic control, our method DTPA that includes all components achieves the best attribute control performance, consistent with the results on sentiment control in Section 5.5. For detoxification, however, we observe that the variant without any components performs best in attribute

#	Pre-Aug	Pro-Aug	ADR	Acc	PPL↓	Dist-1	Dist-2	Dist-3
1	✗	✗	✗	96.43	36.92	0.08	**0.52**	**0.80**
2	✗	✗	✓	97.60	**31.77**	0.08	0.47	0.74
3	✓	✗	✓	97.45	31.84	0.07	0.47	0.74
4	✗	✓	✓	97.73	38.73	0.07	0.47	0.74
5	✓	✓	✓	**98.15**	39.42	**0.08**	0.47	0.74

Table 12: The effects of different components on topic control.

#	Pre-Aug	Pro-Aug	ADR	Tox.↓	PPL↓	Dist-1	Dist-2	Dist-3
1	✗	✗	✗	**15.8**	60.51	**0.08**	**0.49**	**0.80**
2	✗	✗	✓	18.5	**48.34**	0.07	0.44	0.74
3	✓	✗	✓	18.4	48.45	0.07	0.44	0.74
4	✗	✓	✓	20.5	53.52	0.07	0.43	0.74
5	✓	✓	✓	19.7	53.38	0.07	0.43	0.74

Table 13: The effects of different components on detoxification.

control, followed by a significant performance drop after incorporating attribute distribution reconstruction. We hypothesize that this is due to a relatively small gap between the weights of attribute-related and non-attribute-related words in the attribute distribution for this task, and the reconstruction process further narrows this gap, thereby weakening the enhancement of probabilities for attribute-related words. A deeper reason may lie in the limited representational capacity of the trained soft prefixes, which fail to effectively distinguish between attribute-related and non-attribute-related words within the attribute distribution. Prefixes trained on larger and more diverse datasets may acquire stronger representational and guiding capabilities, potentially better highlighting the advantages of our proposed method.

J.2 The impact of Attention Scaling Exponent α

Table 14 and Table 15 show the impact of the attention scaling exponent α on the performance of DTPA in topic control and detoxification, respectively. Overall, both overly large and overly small values of α tend to degrade performance, which has been observed in Section 5.5. Based on these observations, we select the value of α that yields the best attribute control performance for each task. Moreover, by adjusting α, one can flexibly realize personalized controllable text generation. Notably, since α serves as an exponent to modulate the degree of attention amplification, we only consider a few commonly used exponential values (e.g., 1, $1/2$, $1/3, 1/4$). Finer-grained tuning of α may potentially lead to better results.

K Other Results of DTPA in Long Text Generation

In this section, we present the comparison results of DTPA and Air-Decoding in long text generation on topic control and detoxification, as illustrated in Table 16 and 17, respectively. Similar to the results of sentiment control, our method demonstrates consistent superiority on both topic control and detoxification. On topic control, as the sequence length increases, our method shows a more significant improvement in attribute control compared to Air-Decoding. While on detoxification the improvement is not evident, which may be restricted by the relatively weaker attribute control capability of the trained soft prefixes. In terms of fluency, DTPA exhibits a slight decline, due to the increased focus on at-

α	Acc	PPL↓	Dist-1	Dist-2	Dist-3
1	97.40	71.43	0.13	0.55	0.78
$\mathbf{1/2}$	**98.07**	38.03	0.13	0.57	**0.80**
$1/3$	97.07	33.75	**0.14**	**0.58**	0.80
$1/4$	97.27	**32.30**	0.14	0.58	0.79

Table 14: Performance of DTPA under different attention scaling exponents α on topic control.

α	Tox.↓	PPL↓	Dist-1	Dist-2	Dist-3
1	18.8	49.32	0.07	0.43	0.74
$1/2$	18.6	**48.04**	0.07	0.44	0.74
$\mathbf{1/3}$	**18.4**	48.45	**0.07**	**0.44**	**0.74**
$1/4$	18.7	48.67	0.07	0.44	0.74

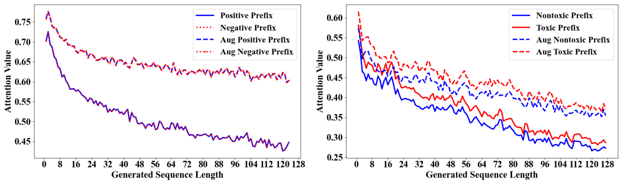
Table 15: Performance of DTPA under different attention scaling exponents α on detoxification.

Length	Method	Acc	PPL↓	Dist-1	Dist-2	Dist-3
64	Air-Decoding	96.23	**28.84**	0.07	0.46	0.75
	DTPA	**97.05**	36.88	**0.07**	**0.46**	**0.76**
128	Air-Decoding	91.98	**24.51**	0.05	**0.43**	0.76
	DTPA	**93.83**	33.59	**0.05**	0.42	**0.77**
192	Air-Decoding	89.03	**23.45**	0.04	0.39	0.76
	DTPA	**90.98**	33.16	**0.04**	**0.39**	**0.76**
256	Air-Decoding	86.95	**22.87**	0.03	**0.37**	**0.75**
	DTPA	**89.10**	32.97	**0.03**	0.36	0.74

Table 16: Performance comparison between DTPA and Air-Decoding under different maximum generation lengths on topic control.

Length	Method	Tox.↓	PPL↓	Dist-1	Dist-2	Dist-3
64	Air-Decoding	19.5	41.26	0.06	0.43	0.75
	DTPA	**19.4**	**41.17**	**0.06**	**0.43**	**0.75**
128	Air-Decoding	23.0	29.98	0.04	0.39	0.74
	DTPA	**22.9**	**29.87**	**0.04**	**0.39**	**0.74**
192	Air-Decoding	23.2	**26.60**	0.04	0.36	0.72
	DTPA	**23.1**	26.70	**0.04**	**0.36**	**0.73**
256	Air-Decoding	23.6	**25.20**	0.03	0.34	0.71
	DTPA	**23.5**	25.33	**0.03**	**0.34**	**0.71**

Table 17: Performance comparison between DTPA and Air-Decoding under different maximum generation lengths on detoxification.



(a) Sentiment Control (b) Detoxification

Figure 6: Visualization of prefix attention augmentation on topic control on (a) sentiment control and (b) detoxification.

tributes suppressing attention to the context. For diversity, DTPA obtains results comparable to Air-Decoding, highlighting its ability to enhance attribute control without sacrificing generation diversity.

L More Visualization Results of Prefix Attention Augmentation

In this section, we present the visualization results of prefix attention augmentation on sentiment control and detoxification, as shown in Figure 6(a) and Figure 6(b), respectively. It can be observed that DTPA significantly alleviate the prefix attention decay across these tasks, enabling the attribute distributions to focus more on attribute-related words, thereby increasing their weights and mitigating the degradation of attribute control performance. Moreover, we can find that the magnitude of prefix attention augmentation on detoxification is smaller than that on sentiment control, corresponding to the smaller attention scaling exponent α ($1/3$ vs. $1/2$). Notably, despite the use of hard prefixes in sentiment control, DTPA still enhance prefix attention effectively, indicating our attention augmentation is insensitive to the choice of prefix type.

M Case Studies

Table 18 presents a comparison of generated examples (maximum generated length is 50) between DTPA and Air-Decoding across different attribute settings. Overall, DTPA generates texts that better satisfy the target attributes while maintaining a reasonable balance of fluency and diversity. However, some outputs still contain grammatical and commonsense errors. This may be attributed to two factors: (1) the strong emphasis on attribute-related words may suppress the selection of non-attribute words, and (2) the limitations of GPT-2's training corpus and reasoning capabilities, which can lead to factual inconsistencies in domains such as science and world-related topics.

Tar. Att	Method	Generated text
Positive	Air-Decoding	**The movie** is based on the true story of Howard A. Brown, a junior the year after President William McKinsey was assassinated during the Spanish-American War. In 1931 he returned to Vietnam and became the first U.S. resident Vietnamese in the
	DTPA	**The movie** looks amazing! The pace seems great and the characters seem to relax together in a way that remains true to the book. I loved that this was a movie that took me back to the good old days of 1980s classic cartoons! They're
Negative	Air-Decoding	**The movie** doesn: 1. Attempt to be funny 2. Cut out the poop 3. Use inane things like dialogue, plot, and dialogue interjections 4. Use dialogue that even the characters think is meta though you already acknowledged it's funny
	DTPA	**The movie** lacks balance and atmosphere\nThere is no character development whatsoever for this story. The following scenes are horribly overwrought and you have a movie about a woman who is a bully who will murder anyone who dares do so, even if they are
World	Air-Decoding	**The relationship** with Israel deteriorated after Hamas formally accepted Jerusalem as Israel's capital in 1994 after a year of violent violence. Hamas leader Khaled Meshaal was among its leaders.\n\nImage copyright AFP Image caption Egyptian President Abdel Fatah al-S
	DTPA	**The relationship** with Israel deteriorated sharply after Hamas seized Gaza from Fatah in August 2011. Demonstrators were arrested and subsequently executed in public squares by the military governorate's security forces.
Sports	Air-Decoding	**More importantly,** athletes like this are the athletes that I fell in love with. There is a person that stood before me like that jersey was going to crack, and beyond any doubt his dedication to his sport was as fierce as it had ever been in
	DTPA	**More importantly,** Durant has become one of the best players in the NBA.\n\n\nOnce the Thunder have developed a better understanding of which positions exist for Durant, Russell Westbrook also will. A lot.\n\n\nNow look around the Western Conference
Business	Air-Decoding	**To conclude,** we looked at earnings reports over the past few years. If stocks were expected earnings growth, we believe the overall price level would be higher due to rising oil production.\n\nGasoline, however, is not the only bright spot in
	DTPA	**To conclude,** sales data for January show a rebound in autos and autos-related services including services for businesses are expected for the first time since December.\n\n\nOctober was another solid month for autos with sales topping $1.104 billion and the
Science	Air-Decoding	**Prior to this** discovery, scientists predicted that sharks should have larger eyes but smaller eyes. These new observations also indicate that the inner edges of dorsal plates can be larger and wider than previously thought. This in turn indicates that the area has an inner ocell
	DTPA	**Prior to this** study, scientists thought that microbes weren't capable of self-assembling into complex matter-like organic molecules. The finding by the authors that they can use this high-level level of chemical innovation to build up complex tissues was intriguing
Nontoxic	Air-Decoding	**I dont now for you, guys, but I** have read the article a few times and tried to understand it.\n\nI believe that (1) only a single-system approach that focuses on minimizing or eliminating dependency (perception, intent, implementation) is required (otherwise the system would
	DTPA	**I dont now for you, guys, but I** have read the article and agree that it is missing important background data or information relevant to the development process. I was sure that it would not have been possible to add it, even with this.\n\nIt seems quite possible, given what is missing

Table 18: Comparison of generated examples by DTPA and Air-Decoding across different attribute settings. "Tar. Att." in the first column represents the target attribute. The given prompts are in bold.